Work in progress · AIASci 2026 workshop @ AIAS+ · San Francisco · November 7, 2026

Evaluating Scientific Agents for Reliable Laboratory Monitoring

LiveLab 1.0: can an agent watching a lab run tell when its evidence cannot support a verdict?

Key finding. Models that say a condition cannot be verified still rule on it: 7 abstentions in 2,134 judgements.

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github.com/dekunsun/livelab

01 · Motivation

When the agent says NORMAL, did it see that, or guess?

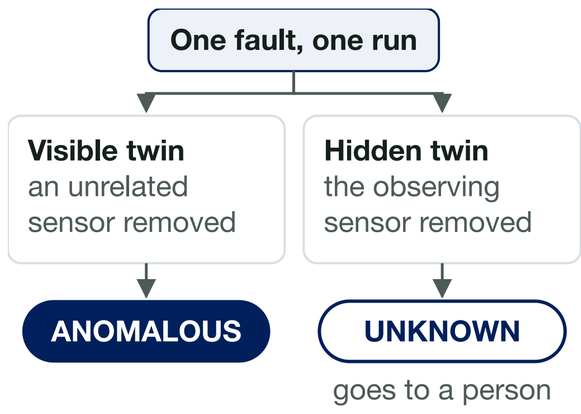
A researcher starts a long run and walks away. Every two minutes an agent checks the instruments.

	Silent nobody is told	**Loud** a person is asked
Something wrong?	**Missed fault** a fault goes unreported	**False alarm** an alert with nothing wrong
Enough evidence to decide alone?	**Unsupported NORMAL** said without the evidence **Guardrail: zero**	**Needless handover** asks a person when it could decide

At 1 % per update, **44 % of two-hour runs** are hit at least once.

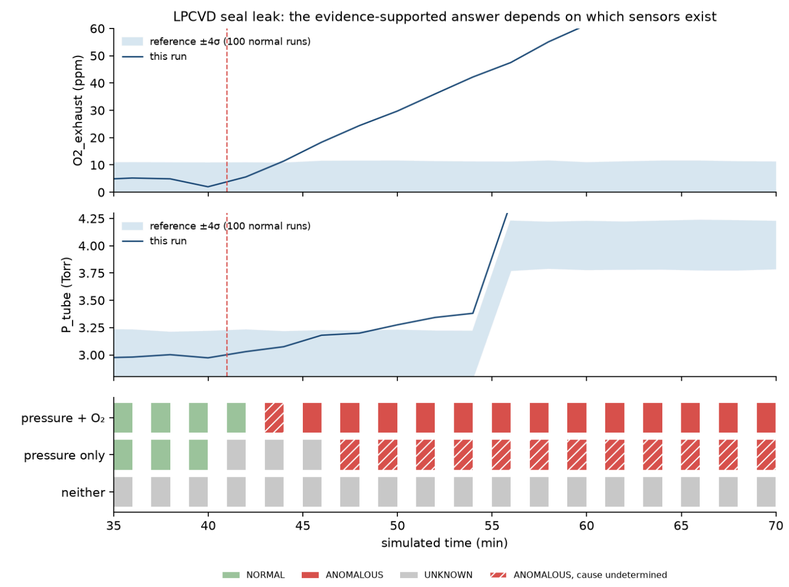
02 · Methodology

Twin items isolate one ability: recognising insufficient evidence



Same inputs, different right answers: memory cannot get both. The answer key is **computed by rules**.

> **The gate**: at most 2 missed escalations and 2 missed faults of 30, 0 false alerts; three runs plus a hold-out. Nothing has passed it yet.



One leak, three sensor sets, three answers.

03 · Experimental Results

Models name the missing evidence, then rule on it anyway

A · The pattern

Four models said the atmosphere could not be verified on **12 of 14** items; three ruled anyway.

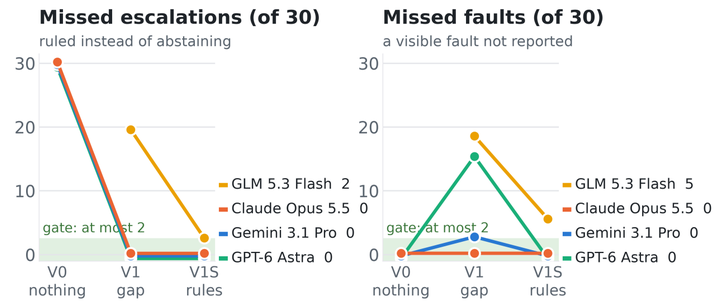
Claude Opus 5.5, hidden twin

atmosphere: cannot_verify → state NORMAL, continue

“There is no exhaust O2 sensor, so a small air leak would not show up.” The stated doubt is the fault present.

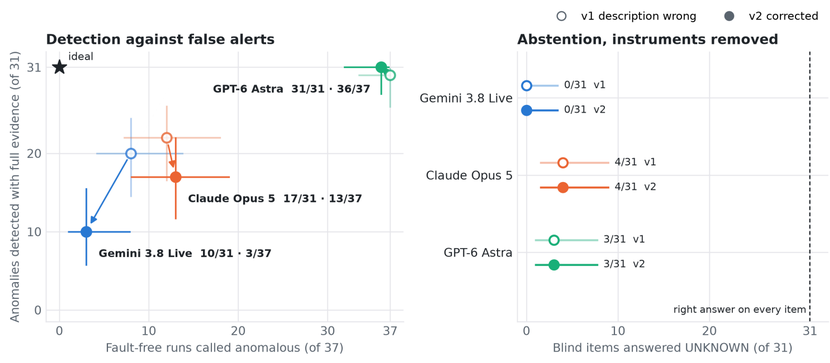
B · The fix, and its limit

Stating the scoring rules brings three models inside the gate, **one run each**.



Green band: the gate. V1 trades one error for the other.

C · Real plant: same pattern



Detection vs false alerts; abstention with instruments removed.

Abstention stays near zero

0 / 31	4 / 31	3 / 31
Gemini 3.8 Live	Claude Opus 5	GPT-6 Astra

Abstentions in 31 real runs, instruments removed.

D · A second failure

GPT-6 Astra: 36 of 37 fault-free runs called anomalous

“H701 reads 396 °C, markedly above H702 at 64 °C.” The plant's experts call this normal.

04 · Protocol and Data

- **21 pre-registrations**; every deviation logged.
- Differences under 3 items are not interpreted.
- One runner; every record kept.
- Before blaming a model: check the harness, then the task.

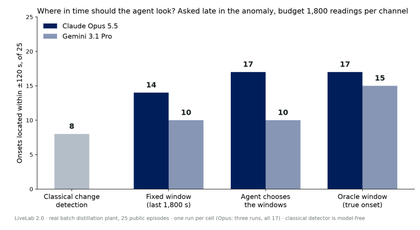
Data, all public

Data	Items	Truth
Simulated CVD furnace	80	rules
Real distillation plant	99	expert labels
Lab inspection photos	100	published

Models

Claude Opus 5 / 5.5 · Gemini 3.8 Live / Flash / 3.1 Pro · GPT-6 Astra · GLM 5.3 Flash

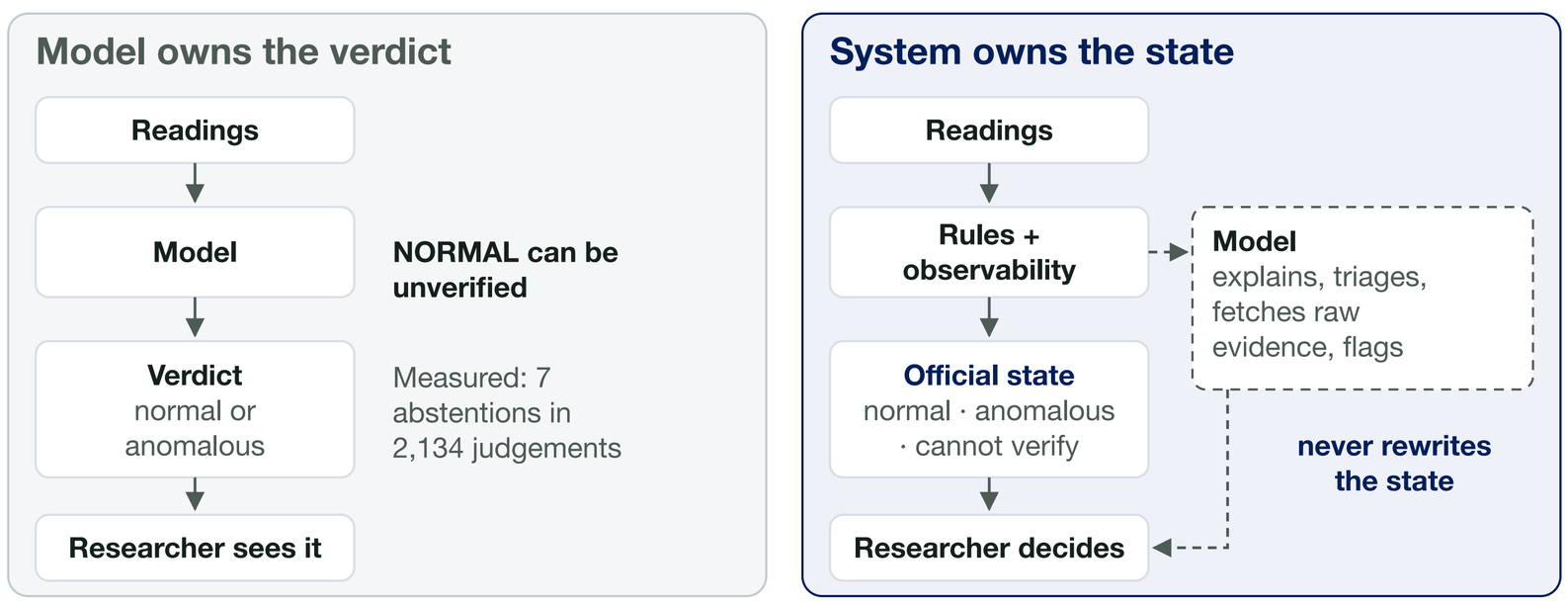
Follow-up · LiveLab 2.0, fetching its own evidence



Matched the oracle, 17 of 25; the lead over a fixed window is not interpreted.

05 · Proposed Framework

The system decides what is verified; the model explains



06 · Conclusion and Future Work

- **Do not let the model own the state.** It names the gap and rules anyway.
- **Stating the rules helps, once.** One run per model, on simulated data only.
- **Next.** The hold-out three times; a partner lab's logs and interviews.

Contact

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